

Computational Linguistics: Final Project Proposal

Wendy Shi

In the study, I will calculate 10 words more likely to occur with *China*, *Russia*, and *Japan* in news related to *technology*, *economy*, and *politics* using Word2Vec word embedding methods.

In this research, I am interested in exploring the geopolitical relationship between China and the US in different news contexts. In addition, I am interested in comparing how the US relationship with China differs from other countries such as Japan and Russia. The US commonly perceives Russia as a political and technological rivalry since the Cold War. On the other hand, the US and Japan have deep collaboration in many fields, such as *economy*, *technology*, and military, due to the post-war alliance. The relationship between China and the US stands somewhere in the middle. I am interested in exploring how geopolitical relationships differ by studying how the US perceives these countries differently.

I will acquire my data using the New York Times API to generate all news data available in the past five years containing *China*, *Russia*, and *Japan* in news filters by *technology*, *economy*, and *politics* categories.

I will apply the Word2Vec word-embedding methods to generate the 10 closest words in a window of ten. Since the corpus I generate using the API will likely be short, I am not planning to train the Word2Vec model with my data. I will use my corpus to tune the pre-trained model with the Google News dataset. Since the pre-trained model is also trained on news articles, the default model also gives us insights into how perceptions change on a temporal level.

Below are the three corpus I will use to fine-tune the pre-trained model

- News containing either *Russia*, *Japan*, or *China* in the field of Politics in the past five years

- News containing either *Russia*, *Japan*, or *China* in the field of technology in the past five years
- News containing either *Russia*, *Japan*, or *China* in the field of economy in the past five years

For each fine-tuned model, I will generate three visualizations for the keywords *Russia*, *Japan*, or *China* with the 10 top words in a 10-word window. In the success model, we are expected to see words with vastly different sentiments for Russia and Japan.

If the model ran successfully, the top 10 words for China may appear to be more similar to Russia than Japan in technology. On the other hand, the top 10 words for China may appear to be more similar to Japan than to Russia in the field of economy. In addition, we will see the context words for China, Russia, and Japan differ, which reflects how the US news media perceive them. If overlapped words exist for different countries in the same field, for example, *AI*. I may also fine-tune country-specific models (E.G. tech news for China, tech news for Japan) that yield the context words for overlapped words.

Word embedding allows us to take a small window around words, which adds more nuisance to semantic analysis. With this study, I hope to gain more insight into how the US perceives countries in different contexts.